BIAP Global - independent verification of stored-evidence reuse (commit 9a6c755, PR #52)

Run at 2026-10-16T15:47:32+00:00 UTC. Reviewer: Claude (Sonnet 5.5). Scope: read-only review of 9a6c755 plus non-destructive regression on an **isolated copy** of the stored data. Nothing merged, deployed, deleted, restarted or bulk-downloaded; production `/var/lib/biap-global` was not read or written.

Conclusion

GO (restricted) for US (NASDAQ, NYSE), GB, FR, IT, SE, NO. NO-GO for DE (XETRA, FRANKFURT) benchmark continuity. Merge of PR #52 is safe for all markets (every failure mode fails closed; defaults unchanged). Deployment still needs owner approval and one live quote read per market (the single online input - see Limits).

Review of 9a6c755

- CI for the exact commit: `global-backend` SUCCESS, `backend` SUCCESS, `android-bundle` SUCCESS, `android-installable-apk` SKIPPED (runs 38917460086 / 38917459908). Full suite re-run locally on the commit: **1550 passed** (matches CI).
- Architecture confirmed: the 24 h `deepResults` copy is derived from the analysis ledger; `evidence_store.rebuild_from_ledger` reads the ledger with `mode=ro`, stamps each row with its original `analyzed_at` (`inputsAsOf`), and needs only current quotes. Gates fail closed (cache left byte-identical).
- Finding (low, non-blocking): `freshness.decision_freshness()` and `classify_filing()` are **not called by any production decision path** (only tests and `ttl_matrix.py`); `quote_state` is attached as metadata (`price_state`) only. The 'BUY needs every critical component' guarantee in practice comes from the pre-existing evidence/governance gates plus `_critical_data_gaps`. It holds empirically (tables below), but the new function is not the enforcement point - wire it in or document it as advisory.
- Finding (cosmetic): `ttl_matrix.py` computes `legacy_expired` from `(base+25h)-base`, which is always >24 h; the 'UNAVAILABLE (24h rule)' column is therefore unconditional (intended as 'at +25 h', but written tautologically). It also hard-codes `/home/ubuntu/...` paths.
- Finding (limit): a rebuilt benchmark stays usable at +25 h but not at +7 d without new quotes/targeted peer rebuild (quote tolerance 4 d; `filing_may_have_changed` for most peers). This is by design.

Method

- Data: copy of `biap-validation-20261016/data-u9` (isolated 14:2x UTC snapshot): analysis ledgers copied; fundamentals/market/shares/filings symlinked read-only; legacy scan caches from `orig-scancache` (DE from data-u9).
- **Quotes were an offline fixture** (stored price x 1.0, dated today) - no network, no bulk download. This proves the reconstruction logic and gates, not the live quote feed.
- Decisions: every ledger `BUY_CANDIDATE` was re-decided offline through the real path (`_derive_metrics` -> core+advanced agents -> evidence -> distress -> governance) from its full stored analysis, with the peer benchmark re-derived from the cache under test.
- Stored-evidence digest before/after: files `['f9743be9df369942', 29123]` == `['f9743be9df369942', 29123]`; per-market ledger content digest and inventory unchanged for all 9 markets.

Per-country result

Market	Stored companies (ledger rows)	Complete	Reusable validated evidence	Complete but missing critical input	Benchmark reconstruction	Rows kept / excluded	Usable at +25 h
DE:XETRA	405	389	21	368	**FAIL (fail-closed)** (REJECTED)	-	False
DE:FRANKFURT	491	456	14	442	**FAIL (fail-closed)** (REJECTED)	-	False
US:NASDAQ	3003	2809	2660	149	**PASS** (OK)	143 / 7	True
US:NYSE	1786	1600	1456	144	**PASS** (OK)	139 / 11	True
GB:LSE	521	420	420	0	**PASS** (OK)	132 / 18	True
FR:EURONEXT_PARIS	293	259	259	0	**PASS** (OK)	124 / 26	True
IT:EURONEXT_MILAN	193	169	169	0	**PASS** (OK)	142 / 8	True
SE:NASDAQ_STOCKHOLM	402	373	373	0	**PASS** (OK)	144 / 6	True
NO:EURONEXT_OSLO	146	136	136	0	**PASS** (OK)	94 / 52	True

Reusable validated = status COMPLETE, evidence PASS, with price, price date, filing period end, sector and earnings (eps or net income) all present in the stored analysis.

Missing critical inputs among COMPLETE rows

Market	filing_period_end	sector	price	price date	earnings	market_cap (info)	Why reconstruction failed (if FAIL)
DE:XETRA	344	240	0	0	2	26	fundamentals not valid for repricing: 79% of peers excluded (possibly superseded/ undated/unvalidated) > 15%; rebuild the excluded peers (see filingDimension.excluded) first
DE:FRANKFURT	408	323	0	0	2	49	fundamentals not valid for repricing: 74% of peers excluded (possibly superseded/ undated/unvalidated) > 15%; rebuild the excluded peers (see filingDimension.excluded) first
US:NASDAQ	0	148	0	0	1	683	
US:NYSE	0	144	0	0	0	602	
GB:LSE	0	0	0	0	0	28	
FR:EURONEXT_PARIS	0	0	0	0	0	182	
IT:EURONEXT_MILAN	0	0	0	0	0	8	
SE:NASDAQ_STOCKHOLM	0	0	0	0	0	70	
NO:EURONEXT_OSLO	0	0	0	0	0	131	

Expired benchmark -> reconstruction (no market re-download)

Market	Legacy cache age now	Legacy usable under 24 h rule at +25 h	Reconstructed from ledger	Quote source	Dry run wrote nothing	Ledger+inventory unchanged
DE:XETRA	5.8 h	No	REJECTED (None peers)	offline fixture	True	True
DE:FRANKFURT	92.4 h	No	REJECTED (None peers)	offline fixture	True	True
US:NASDAQ	18.1 h	No	OK (143 peers)	offline fixture	True	True
US:NYSE	18.1 h	No	OK (139 peers)	offline fixture	True	True
GB:LSE	18.1 h	No	OK (132 peers)	offline fixture	True	True
FR:EURONEXT_PARIS	18.0 h	No	OK (124 peers)	offline fixture	True	True
IT:EURONEXT_MILAN	18.0 h	No	OK (142 peers)	offline fixture	True	True
SE:NASDAQ_STOCKHOLM	18.0 h	No	OK (144 peers)	offline fixture	True	True
NO:EURONEXT_OSLO	18.0 h	No	OK (94 peers)	offline fixture	True	True

Actual decision results (ledger BUY_CANDIDATEs re-decided offline)

Market	Ledger BUYs	Original <24 h benchmark	Rebuilt benchmark	Expired benchmark (no rebuild)	New BUYs created by rebuild
DE:XETRA	0	-	-	-	0
DE:FRANKFURT	1	NO_RECOMMENDATION 1	NO_RECOMMENDATION 1	NO_RECOMMENDATION 1	0
US:NASDAQ	19	BUY_CANDIDATE 6, HOLD_OR_WATCH 4, NO_RECOMMENDATION 9	BUY_CANDIDATE 6, HOLD_OR_WATCH 4, NO_RECOMMENDATION 9	HOLD_OR_WATCH 2, NO_RECOMMENDATION 17	0
US:NYSE	18	BUY_CANDIDATE 10, HOLD_OR_WATCH 2, NO_RECOMMENDATION 6	BUY_CANDIDATE 10, HOLD_OR_WATCH 2, NO_RECOMMENDATION 6	HOLD_OR_WATCH 3, NO_RECOMMENDATION 15	0
GB:LSE	17	BUY_CANDIDATE 10, HOLD_OR_WATCH 5, NO_RECOMMENDATION 2	BUY_CANDIDATE 7, HOLD_OR_WATCH 5, NO_RECOMMENDATION 5	HOLD_OR_WATCH 6, NO_RECOMMENDATION 11	0
FR:EURONEXT_PARIS	3	HOLD_OR_WATCH 1, NO_RECOMMENDATION 2	HOLD_OR_WATCH 1, NO_RECOMMENDATION 2	HOLD_OR_WATCH 1, NO_RECOMMENDATION 2	0
IT:EURONEXT_MILAN	5	BUY_CANDIDATE 3, NO_RECOMMENDATION 2	BUY_CANDIDATE 3, NO_RECOMMENDATION 2	HOLD_OR_WATCH 2, NO_RECOMMENDATION 3	0
SE:NASDAQ_STOCKHOLM	1	HOLD_OR_WATCH 1	HOLD_OR_WATCH 1	NO_RECOMMENDATION 1	0
NO:EURONEXT_OSLO	10	BUY_CANDIDATE 1, HOLD_OR_WATCH 1, NO_RECOMMENDATION 8	BUY_CANDIDATE 1, HOLD_OR_WATCH 1, NO_RECOMMENDATION 8	HOLD_OR_WATCH 3, NO_RECOMMENDATION 7	0

Rebuilt decisions are identical to the original-benchmark control in every market except GB LSE. GB LSE: 7 of 10 BUYs retained (3 became HOLD/NO_RECOMMENDATION because 18 peers lacking a valid multiple/quote or possibly-superseded filing were excluded - conservative). Rebuild never created a BUY that the original benchmark did not (0 in all markets). With the benchmark simply expired, **0 BUY_CANDIDATE in every market** (fail closed).

Missing critical evidence cannot produce BUY_CANDIDATE

Each ledger BUY was re-decided with one critical input removed (rebuilt benchmark in place). Cell = number of resulting BUY_CANDIDATE out of BUYs tested.

Market	no price	no earnings	no filing period	no sector	price 40 d old
DE:XETRA	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0
DE:FRANKFURT	0 / 1	0 / 1	0 / 1	0 / 1	0 / 1
US:NASDAQ	0 / 19	0 / 19	0 / 19	0 / 19	0 / 19
US:NYSE	0 / 18	0 / 18	0 / 18	0 / 18	0 / 18
GB:LSE	0 / 17	0 / 17	0 / 17	0 / 17	0 / 17
FR:EURONEXT_PARIS	0 / 3	0 / 3	0 / 3	0 / 3	0 / 3
IT:EURONEXT_MILAN	0 / 5	0 / 5	0 / 5	0 / 5	0 / 5
SE:NASDAQ_STOCKHOLM	0 / 1	0 / 1	0 / 1	0 / 1	0 / 1
NO:EURONEXT_OSLO	0 / 10	0 / 10	0 / 10	0 / 10	0 / 10

A first mutation set that cleared only `eps/pe` still yielded BUYs because P/E is re-derived from net income and market cap; clearing all earnings fields yields 0. That was test design, not a defect.

Stale prices are never presented as current; history stays accessible

Market	All quotes 10 d old	10% of quotes 10 d old	Stale rows keeping multiples	Stale rows labelled current	inputsAsOf == ledger analyzed_at	Fundamentals sample readable (history files)
DE:XETRA	REJECTED (0 rows repriced)	REJECTED	0	0	0/150	25/25 (47)
DE:FRANKFURT	REJECTED (0 rows repriced)	REJECTED	0	0	0/150	25/25 (46)
US:NASDAQ	REJECTED (0 rows repriced)	REJECTED	0	0	150/150	25/25 (26)
US:NYSE	REJECTED (0 rows repriced)	OK (15 excluded)	0	0	150/150	25/25 (25)
GB:LSE	REJECTED (0 rows repriced)	REJECTED	0	0	150/150	25/25 (25)
FR:EURONEXT_PARIS	REJECTED (0 rows repriced)	REJECTED	0	0	150/150	25/25 (30)
IT:EURONEXT_MILAN	REJECTED (0 rows repriced)	OK (15 excluded)	0	0	150/150	25/25 (26)
SE:NASDAQ_STOCKHOLM	REJECTED (0 rows repriced)	OK (15 excluded)	0	0	150/150	25/25 (49)

Market	All quotes 10 d old	10% of quotes 10 d old	Stale rows keeping multiples	Stale rows labelled current	inputsAsOf == ledger analyzed_at	Fundamentals sample readable (history files)
NO:EURONEXT_OSLO	REJECTED (0 rows repriced)	REJECTED	0	0	146/146	25/25 (36)

Stale quotes either reject the whole rebuild (cache left byte-identical) or exclude the row with its multiples nulled; no row is relabelled current and `cachedAt` is never rewritten. Stored ledger rows keep their original analysis timestamp; archived fundamentals are intact (digest identical).

Germany

DE cannot be reconstructed from its ledger: `filing_period_end` is missing in 344 of 389 COMPLETE XETRA rows (408 of 456 FRANKFURT) and sector in 240 / 323, so 79% / 74% of peers are excluded (> 15% gate). The gate fails closed: XETRA's legacy cache (5.8 h old) will expire at 24 h with no replacement, and FRANKFURT's is already 92 h old. No DE BUY exists today (XETRA 0, FRANKFURT 1 -> NO_RECOMMENDATION), so no unsafe decision results, but benchmark continuity for DE needs a targeted peer rebuild (period-end extraction) - already listed as open in the PR.

Limits and what this does not prove

- Live quote retrieval was not exercised (offline fixture by instruction: no bulk downloads); earlier live runs are in the PR's TTL matrix.
- Re-decision is offline from stored analyses and is not byte-identical to production scoring; the control column (original benchmark, same harness) is the fair comparison.
- Per-market rebuild uses the top-150 ledger rows (scanner cap), BUY/high-score first.
- Nothing was deployed; production caches still follow the legacy 24 h rule until an approved deploy + `rebuild`/`reprice` run.